



School of Informatics & IT
TEMASEK POLYTECHNIC

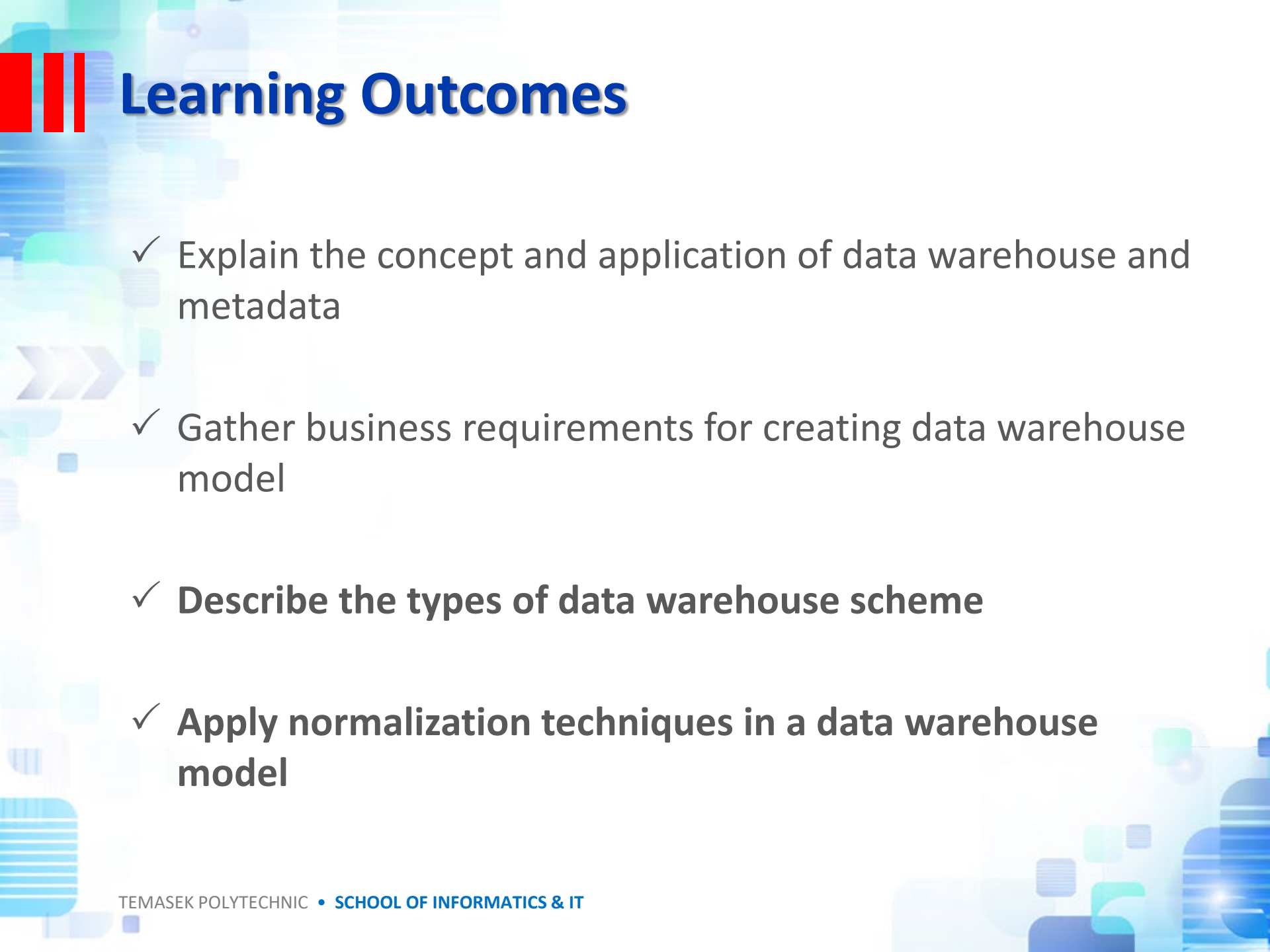
Data Warehousing and Business Intelligence

CDA2C01



Topic 2

Introduction to Data Warehouse Model Design IV



Learning Outcomes

- ✓ Explain the concept and application of data warehouse and metadata
- ✓ Gather business requirements for creating data warehouse model
- ✓ **Describe the types of data warehouse scheme**
- ✓ **Apply normalization techniques in a data warehouse model**



SLOWLY CHANGING DIMENSION

Type 0

Data in Operational Database: Customer table

CustomerID	Name	State
1001	Christina	Illinois

Change to

CustomerID	Name	State
1001	Christina	Texas

Data in DW: Customer Dimension

CustomerKey	CustomerID	Name	State
1	1001	Christina	Illinois

In Type 0 Slowly Changing Dimension, no new information will be updated.

Data in warehouse will remain unchanged

Type 1

Data in Operational Database: Customer table

CustomerID	Name	State
1001	Christina	Illinois

Change to

CustomerID	Name	State
1001	Christina	Texas

Data in DW: Customer Dimension

CustomerKey	CustomerID	Name	State
1	1001	Christina	Texas

In Type 1 Slowly Changing Dimension, the new information simply **overwrites** the original information. In other words, no history is kept.

Type 2

Data in Operational Database: Customer table

CustomerID	Name	State
1001	Christina	Illinois

Change to

CustomerID	Name	State
1001	Christina	Texas

Data in DW: Customer Dimension

CustomerKey	Effective	Expired	Current	CustomerID	Name	State
1	2018-10-01	2020-09-30	0	1001	Christina	Illinois
2	2020-10-01	Null	1	1001	Christina	Texas

In Type 2 Slowly Changing Dimension, a new **record** is added to the table to represent the new information. Therefore, both the original and the new record will be present. The new record gets its own primary key.

Type 3

Data in Operational Database: Customer table

CustomerID	Name	State
1001	Christina	Illinois



CustomerID	Name	State
1001	Christina	Texas

Data in DW: Customer Dimension

CustomerKey	Effective	CustomerID	Name	Original_state	Current_state
1	2018-10-01	1001	Christina	Illinois	Texas

In Type 3 Slowly Changing Dimension, there will be two columns to indicate the particular attribute of interest, one indicating the original value, and one **additional** column indicating the current value. There will also be a column that indicates when the current value becomes active.

Type 6: Hybrid: Type 1 +2 +3

Data in Operational Database: Customer table

CustomerID	Name	State
1001	Christina	Illinois



CustomerID	Name	State
1001	Christina	Texas

Data in DW: Customer Dimension

CustomerKey	Effective	Expired	Current	CustomerID	Name	Historical_State	Current_State
1	2018-10-01	2020-09-30	0	1001	Christina	Illinois	Texas
2	2020-10-01	Null	1	1001	Christina	Texas	Texas

For Type 6, we overwrite the “Current Information” in all relevant historical records (Row_Key = 1) with the new information, as in Type 1 processing. We create a new record to track the changes, as in Type 2 processing. And we store the history in a second column which incorporates Type 3 processing. By **design**, it will be easier to analysis using aggregation of both historical and current information.

Group work

- What happens when Christina moves again, from Texas to New York, on 2021-01-01 ?
- Prepare and present your solution for the dimension table as of today, for Type 0, 1, 2, 3 and 6.



NORMALIZATION

What is Normalization?

- Normalization is a database design technique that reduces data redundancy and eliminates undesirable characteristics like Insertion, Update and Deletion Anomalies.
- Normalization rules divides larger tables into smaller tables and links them using relationships. The purpose of Normalization in SQL is to eliminate redundant (repetitive) data and ensure data is stored logically.

Database Normal Forms

- Here is a list of Normal Forms
 - **1NF (First Normal Form)**
 - **2NF (Second Normal Form)**
 - **3NF (Third Normal Form)**
 - **BCNF (Boyce-Codd Normal Form)**
 - **4NF (Fourth Normal Form)**
 - **5NF (Fifth Normal Form)**
 - **6NF (Sixth Normal Form)**

UNF: Unnormalized form

- Without any normalization, all information is stored in one table as shown below.

FULL NAMES	PHYSICAL ADDRESS	MOVIES RENTED	SALUTATION
Janet Jones	First Street Plot No 4	Pirates of the Caribbean, Clash of the Titans	Ms.
Robert Phil	3 rd Street 34	Forgetting Sarah Marshal, Daddy's Little Girls	Mr.
Robert Phil	5 th Avenue	Clash of the Titans	Mr.

1NF (First Normal Form)

- Rules:
 - Each table cell should contain a single value.
 - Each record needs to be unique.

FULL NAMES	PHYSICAL ADDRESS	MOVIES RENTED	SALUTATION
Janet Jones	First Street Plot No 4	Pirates of the Caribbean	Ms.
Janet Jones	First Street Plot No 4	Clash of the Titans	Ms.
Robert Phil	3 rd Street 34	Forgetting Sarah Marshal	Mr.
Robert Phil	3 rd Street 34	Daddy's Little Girls	Mr.
Robert Phil	5 th Avenue	Clash of the Titans	Mr.

2NF (Second Normal Form)

- Rules:
 - Be in 1NF
 - Single Column Primary Key

Which or if all are 2NF?

MEMBERSHIP ID	FULL NAMES	PHYSICAL ADDRESS	SALUTATION
1	Janet Jones	First Street Plot No 4	Ms.
2	Robert Phil	3 rd Street 34	Mr.
3	Robert Phil	5 th Avenue	Mr.

Table 1

MEMBERSHIP ID	MOVIES RENTED
1	Pirates of the Caribbean
1	Clash of the Titans
2	Forgetting Sarah Marshal
2	Daddy's Little Girls
3	Clash of the Titans

Table 2

3NF (Third Normal Form)

- Rules:
 - Be in 2NF
 - Has no transitive functional dependencies

MEMBERSHIP ID	FULL NAMES	PHYSICAL ADDRESS	SALUTATION ID
1	Janet Jones	First Street Plot No 4	2
2	Robert Phil	3 rd Street 34	1
3	Robert Phil	5 th Avenue	1

TABLE 1

MEMBERSHIP ID	MOVIES RENTED
1	Pirates of the Caribbean
1	Clash of the Titans
2	Forgetting Sarah Marshal
2	Daddy's Little Girls
3	Clash of the Titans

Table 2

SALUTATION ID	SALUTATION
1	Mr.
2	Ms.
3	Mrs.
4	Dr.

Table 3

UNF-remove repetition group>1NF
1NF-remove functional dependency>2NF
2NF-remove transitive dependency>3NF

Transitive dependency

Remember
your LOMA?

Let A, B, and C designate three distinct (but not necessarily disjoint) sets of attributes of a relation. Suppose all three of the following conditions hold:

1. $A \rightarrow B$
2. It is not the case that $B \rightarrow A$
3. $B \rightarrow C$

Then the functional dependency $A \rightarrow C$ (which follows from 1 and 3 by the [axiom of transitivity](#)) is a transitive dependency.

Example [\[edit \]](#)

A transitive dependency occurs in the following relation:

Book	Genre	Author	Author Nationality
<i>Twenty Thousand Leagues Under the Sea</i>	Science Fiction	Jules Verne	French
<i>Journey to the Center of the Earth</i>	Science Fiction	Jules Verne	French
<i>Leaves of Grass</i>	Poetry	Walt Whitman	American
<i>Anna Karenina</i>	Literary Fiction	Leo Tolstoy	Russian
<i>A Confession</i>	Religious Autobiography	Leo Tolstoy	Russian

The functional dependency $\{\text{Book}\} \rightarrow \{\text{Author Nationality}\}$ applies; that is, if we know the book, we know the author's nationality. Furthermore:

- $\{\text{Book}\} \rightarrow \{\text{Author}\}$
- $\{\text{Author}\}$ does not $\rightarrow \{\text{Book}\}$
- $\{\text{Author}\} \rightarrow \{\text{Author Nationality}\}$

Therefore $\{\text{Book}\} \rightarrow \{\text{Author Nationality}\}$ is a transitive dependency.

Transitive dependency occurred because a non-key attribute (Author) was determining another non-key attribute (Author Nationality).